DD-Mamba: A Forecast-Decomposable Dual-Domain State-Space Framework for Multivariate Time-Series Forecasting

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Abstract

Long-term multivariate forecasting requires a model to represent temporal dynamics, periodic structure, and cross-variable dependence without assuming that every dataset benefits from the same capacity. Existing dual-domain models usually fuse these sources internally, making their individual effects difficult to assess. We present DD-Mamba, a forecast-decomposable state-space framework in which a decomposition-based temporal branch and a learned spectral branch each produce a complete forecast and are combined by a convex gate. A switchable bidirectional variate mixer provides cross-channel capacity, while a linear temporal pathway and zero-initialized corrections give the model a simple starting map. The design supports controlled component removal rather than inferring component value from a single end-to-end score. On six benchmarks rerun in one pipeline against seven baselines, DD-Mamba obtains the lowest average MSE on five datasets; seed-matched tests with Benjamini–Hochberg correction support improvements over the strongest in-pipeline competitor on four, while Weather is tied and

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PatchTST is better on ETTm1. Across 83 component comparisons, none of the 29 placement effects survives multiplicity correction at three seeds. Eight corrected branch-removal effects support the temporal branch, chiefly on ETTm data. The only corrected frequency-side effect occurs for Solar at horizon 96, but because that dataset's normalization switch was selected with test feedback, we treat it as exploratory rather than confirmatory. Training-set channel correlation and dispersion statistics provide descriptive hypotheses for capacity placement, not a validated selector. The results therefore support a component-wise analysis framework and bounded dataset-specific findings, not universal superiority or a general rule for model selection.
Keywords: Time series forecasting, State-space models, Mamba, Frequency domain, Multivariate analysis

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1. Introduction

Long-term forecasting of multivariate time series underpins applications from power-grid load planning to traffic management and weather-dependent operations. The field has cycled through architectural paradigms at a remarkable pace: Transformer variants with efficient attention [1, 2, 3], patch-based attention [4], inverted variate-token attention [5], and, most recently, selective state-space models (SSMs) built on Mamba [6, 7, 8], which offer linear-time sequence modeling with input-dependent dynamics.

Combining time- and frequency-domain representations is itself an established direction: TF4TF [9] models multi-semantic time–frequency information for long-term forecasting, and recent work pairs time–frequency or decomposition-based modeling with Mamba backbones [10, 8]. We therefore do *not* position dual-domain forecasting or a time–frequency Mamba as our contribution. What remains under-examined is a different question — *when* does each such component actually pay? Existing dual-domain models fuse representations inside the network, so the time view, the spectral view, normalization, and cross-variate capacity are entangled and cannot be individually switched off and measured. Our contribution is a *forecast-decomposable* framework that makes exactly these choices separable and testable, together with the seed-paired, multiplicity-controlled evidence of where each one helps.

Three empirical gaps motivate this design. **Gap 1: deep models discard strong linear maps.** A per-component linear projection from the look-back window to the horizon (DLinear [11]) and a tiny complex-linear interpolation of the low-passed spectrum (FITS [12]) can be competitive with much larger models on standard benchmarks. Without such a direct pathway, a deep architecture must learn any linear behavior implicitly, which may be inefficient on small datasets. **Gap 2: the useful inductive bias differs by domain, but is rarely separable.** Local trends and regime changes are naturally expressed in the time domain, while multi-scale seasonality is compact in the frequency domain. Models committed to a single view — attention or SSMs over time steps only, or spectral mixing only [13] — expose just one; and even dual-domain models that use both, such as TF4TF [9], fuse them *inside* the network, so neither view is a separately removable forecast path whose marginal value can be measured. **Gap 3: cross-variate capacity can help or hurt.** Cross-variate mixing is effective on several high-dimensional benchmarks [5, 7, 14], but the underlying data property it exploits — how strongly channels are actually correlated — differs by a factor of three across the standard benchmarks (Table 1). This motivates treating the mixer as an explicit option rather than a universal setting, while testing rather than assuming that correlation predicts its value.

We answer these three gaps with DD-Mamba (Fig. 1), mapping each to one structural remedy: *(i)* a DLinear-style time backbone with zero-initialized nonlinear corrections; *(ii)* parallel time and frequency forecasts, including an optional dormant FITS-style spectral map, fused by a convex gate; and *(iii)* a bidirectional variate mixer switched on or off per dataset. Seed-matched ablations test these choices below; effect sizes and confidence intervals are reported for every comparison, and a finding is considered statistically supported only when it survives the stated multiplicity correction.

Concretely, the time branch combines a DLinear-style pathway [11] with a zero-initialized nonlinear correction, while the frequency branch combines learned filtering with an optional zero-initialized FITS-style spectral pathway [12]. The resulting model starts as a linear map of the instance-normalized window (the RLinear-class structure [15, 16]); a controlled test finds zero initialization accuracy-neutral. The branch forecasts form a learned convex combination. Because each branch emits a full forecast, we can remove it and measure the effect with seed-paired tests, correcting for multiple comparisons with the Benjamini–Hochberg (BH) procedure. Two patterns emerge. Component-placement effects — the variate mixer helping Weather while hurting ETTh1, normalization helping Weather while hurting Solar — are visible at the raw level but, at three seeds, *none* survive BH correction; we therefore report them as *exploratory* directional evidence, not confirmed effects. Branch removal is sharper: of 54 branch tests, nine meet the BH threshold. Eight concern the *time* branch on ETTm1, ETTm2, Electricity, and Exchange. The ninth is the frequency-side Solar cell at H=96; however, because Solar's normalization switch was selected after test-set inspection, that result is treated as exploratory. The current clean evidence therefore supports the temporal pathway but does not establish a general dual-domain advantage.

Contributions. **(1)** A *forecast-decomposable* architecture in which temporal and spectral paths emit complete predictions and the normalization, variate mixer, and branch pathways can be removed without redefining the forecasting task. **(2)** A component-level evaluation protocol that reports seed-matched effect sizes and confidence intervals and controls the false discovery rate across 83 comparisons; importantly, it distinguishes the eight supported temporal-branch effects from exploratory placement and frequency-side findings. **(3)** A data-centric analysis relating the measured effects to training-set cross-channel correlation, horizon, and dispersion. These statistics are used to formulate testable hypotheses, not presented as a validated selection rule.

Organization. Section 2 reviews time series analysis tasks, deep models for temporal and for cross-dimensional dependencies, and the resulting research gap. Section 3 formalizes the forecasting problem and reviews the state-space background DD-Mamba builds on. Section 4 details the dual-domain architecture, its zero-initialized corrections, and the switchable variate mixer. Section 5 reports the main comparison, paired ablations, and the branch-ablation matrix; Section 6 distills practical component-selection guidance and threats to validity; and Section 7 concludes.

2. Related Work

2.1. Time Series Analysis Tasks

Deep models now serve the full spectrum of time series mining tasks — forecasting, classification, anomaly detection, and imputation — often through shared general-purpose backbones such as TimesNet [17], whose representations transfer across tasks. Long-term multivariate *forecasting*, our setting, is the most benchmark-standardized of these problems: fixed look-back, multi-horizon evaluation, and shared splits [1, 5] make it a suitable arena for controlled architectural evidence. Although backbone ideas transfer across tasks, the component effects measured here are specific to forecasting and should not be assumed to hold for classification, anomaly detection, or imputation without separate tests.

2.2. Temporal-Dependency Models

One line models dependencies along the time axis only. Transformer variants adapt attention to long sequences via sparsity, decomposition, and frequency enhancement [1, 2, 3], patching [4], or explicit de-/re-stationarization [18]. A second family shows that *linear* maps remain competitive: DLinear/NLinear [11], RLinear with reversible instance normalization [15, 16], the MLP-based TiDE [19] and TimeMixer [20], and the residual-stacked N-BEATS/N-HiTS [21, 22], our closest residual-forecasting precedents. A third works in the spectrum: FreTS [13] learns frequency-domain MLPs and FITS [12] forecasts with one complex-linear interpolation of the low-passed spectrum. A fourth, directly adjacent to us, combines *both* time and frequency views: TF4TF [9] mines multi-semantic time–frequency information for long-term forecasting. Such dual-domain models fuse the two views inside the network, so — unlike the framework here — neither view is a separately removable forecast path. Decomposition-based designs (STL [23], SCINet [24], and the seasonal-trend-dispersion view of STD [25]) make component structure explicit; RevIN, used here as a per-dataset switch, normalizes windows reversibly but neither extracts nor forecasts dispersion, and we claim no STD-style decomposition.

2.3. Cross-Dimensional (Spatial–Temporal) Models

A second line treats the variate dimension as a first-class axis. Crossformer [14] is the classic Transformer of this type: it embeds the series into a two-dimensional patch grid and attends over *both* the temporal and the cross-variate (spatial) dimension. iTransformer [5] inverts the axes and attends over variate tokens only; SOFTS [26] reaches linear-complexity channel interaction through a centralized core, and ModernTCN [27] interleaves temporal and cross-variable convolutions. Selective state-space models bring linear-time scans to the same axes: Mamba [6, 28] over time, and S-Mamba [7], TimeMachine [29], and FMamba [30] over variate tokens, with recent variants adding multi-scale processing (ms-Mamba [8]) and series decomposition (DecMamba [10]). G-Mamba [31] further combines graph-conditioned state-space dynamics with explicit spatial relations, and TimePro [32] builds variable- and time-aware hyper-states to address multi-delay cross-variable effects. DD-Mamba sits in this lineage — causal Mamba over time and bidirectional Mamba over variates and frequency bins — but treats cross-variable capacity as a switchable component whose measured value can be analyzed.

2.4. Research Gap

Dual-domain forecasting [9], time–frequency and decomposition Mamba hybrids [10, 8], and cross-dimensional attention [14, 5] are all, by now, well-populated spaces; a new model in any of them is unlikely to be a contribution on its own. What these lines share is a *methodological* limitation rather than an architectural one. (i) Representations are fused inside the network, so the marginal value of the time view, the spectral view, or a decomposition step is never separately measured. (ii) Cross-dimensional models such as Crossformer apply the same spatial–temporal capacity to every dataset, although channel coupling varies enormously across benchmarks (mean absolute cross-channel Pearson correlation ranges from 0.31 on ETTh1 to 0.91 on Solar; Table 1). (iii) Component-level comparisons, when reported at all, typically rest on a handful of seeds and uncorrected p-values, so it is unclear which claimed effects would survive multiplicity control. The methodological gap we address is therefore *measurement*: existing architectures provide limited evidence about whether an observed gain comes from temporal modeling, spectral modeling, or cross-variable capacity, and component studies rarely account for the number of comparisons being made. The closest works in spirit are TimeRecipe [33] and CombinationTS [34], which decompose forecasters into interchangeable modules and benchmark module-level effectiveness *across* many architectures at large scale—to distill general design recipes or to quantify per-module marginal effectiveness and stability. We do not attempt that cross-architecture generality: we instead audit components *within* a single forecast-decomposable architecture, using per-seed paired tests with Benjamini–Hochberg control and explicitly labelling any switch chosen with test feedback as exploratory—so our claims are bounded to the studied datasets rather than offered as a transferable recipe or a validated selection rule. A formal estimand, multiple backbones, or successful held-out component selection would be needed to raise this audit to the level of those frameworks; we do not claim it here. Concretely, DD-Mamba enables that analysis through independently removable time/frequency forecasts, a switchable variate mixer, and a 27-cell branch-ablation matrix analyzed under Benjamini–Hochberg control. This yields bounded empirical findings on the studied datasets, not a validated rule for predicting component value on unseen data.

3. Preliminaries

3.1. Problem Formulation

Given a look-back window $\mathbf{X} \in \mathbb{R}^{L\times C}$ of L steps and C variates, long-term multivariate forecasting predicts the next H steps $\mathbf{Y} \in \mathbb{R}^{H\times C}$, with H typically in $\{96, 192, 336, 720\}$. Following RevIN [16], we standardize each window per variate, $\tilde{\mathbf{X}} = (\mathbf{X} - \boldsymbol{\mu})/\boldsymbol{\sigma}$, using the window's own per-variate mean $\boldsymbol{\mu}$ and standard deviation $\boldsymbol{\sigma}$, and de-normalize the final forecast by the inverse affine map. Instance normalization is treated as a per-dataset switch rather than a universal default; Section 5.3 shows why. A model is a map $f_\theta : \tilde{\mathbf{X}} \mapsto \mathbf{Y}$ trained to minimize the mean squared error over the horizon.

3.2. State Space Models

A continuous linear state-space model maps an input signal $u(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through an N-dimensional latent state $\mathbf{h}(t) \in \mathbb{R}^N$,

$$\mathbf{h}'(t) = \mathbf{A}\,\mathbf{h}(t) + \mathbf{B}\,u(t), \qquad y(t) = \mathbf{C}^\top \mathbf{h}(t), \tag{1}$$

with $\mathbf{A} \in \mathbb{R}^{N \times N}$, $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{N}$. To operate on a discrete sequence, the system is discretized with a step size $\boldsymbol{\Delta}$ under a zero-order hold, which yields the discrete parameters

$$\bar{\mathbf{A}} = \exp(\boldsymbol{\Delta}\mathbf{A}), \qquad \bar{\mathbf{B}} = (\boldsymbol{\Delta}\mathbf{A})^{-1}(\exp(\boldsymbol{\Delta}\mathbf{A}) - \mathbf{I})\,\boldsymbol{\Delta}\mathbf{B} \;\approx\; \boldsymbol{\Delta}\mathbf{B}, \tag{2}$$

and turns Eq. (1) into the linear recurrence $\mathbf{h}_k = \bar{\mathbf{A}}\,\mathbf{h}_{k-1} + \bar{\mathbf{B}}\,u_k$, $y_k = \mathbf{C}^\top \mathbf{h}_k$. Classical structured SSMs keep $(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C})$ fixed across the sequence, so the recurrence is linear time-invariant and can be evaluated as a global convolution. Mamba [6] makes the model *selective*: the parameters $(\boldsymbol{\Delta}_k, \mathbf{B}_k, \mathbf{C}_k)$ become functions of the input u_k, so the state transition adapts at each step and the model can selectively remember or forget context. Selectivity breaks the convolutional form, but a hardware-aware parallel scan keeps the cost linear in sequence length. DD-Mamba uses this selective recurrence in two roles — a causal scan over time and, optionally, a bidirectional scan over variates — detailed next.

4. Method

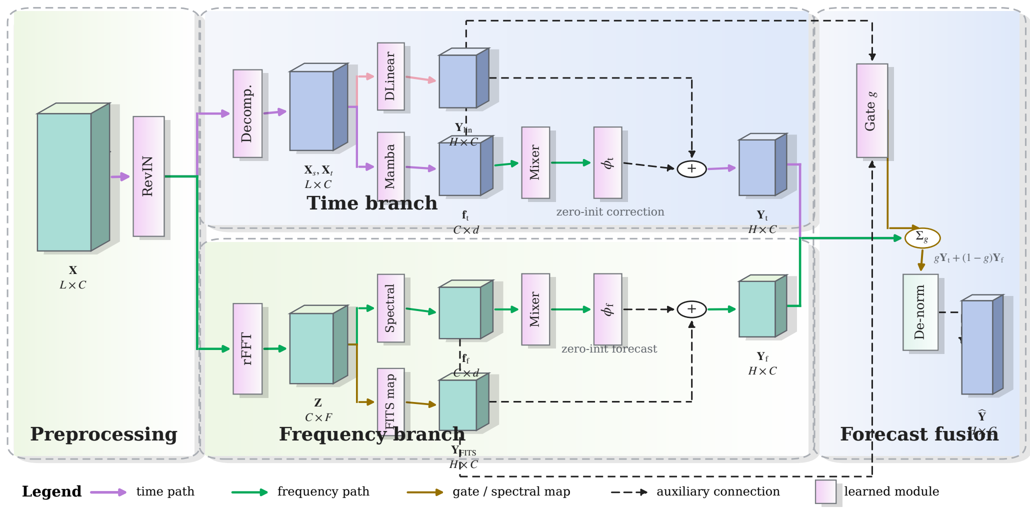

Figure 1: DD-Mamba as a forecast-decomposable architecture. The normalized input is routed to two explicit prediction paths. The time branch combines a DLinear anchor with a causal-Mamba correction; the frequency branch combines a complex spectral encoder with an optional FITS-style linear spectral map. Each branch produces a complete forecast, $\mathbf{Y}_{\mathrm{t}}$ or $\mathbf{Y}_{\mathrm{f}}$, and branch features parameterize the convex gate g. Shadowed cuboids denote intermediate tensors, vertical lavender bars denote learned operators, purple and green arrows trace the time- and frequency-domain paths, gold arrows mark spectral/gating operations, and black dashed arrows denote auxiliary or residual connections. Zero initialization and dataset-level switches are stated beside their associated operators. The final forecast is $g\mathbf{Y}_{\mathrm{t}} + (1-g)\mathbf{Y}_{\mathrm{f}}$, followed by inverse instance normalization.

4.1. Time Branch: Decomposition, Linear Backbone, Selective Scan

The time branch decomposes $\tilde{\mathbf{X}}$ with a moving average into trend $\mathbf{X}_t$ and seasonal $\mathbf{X}_s$ components [2, 11] and forecasts with two cooperating paths.

Linear backbone. Per-component linear maps project the window directly to the horizon:

$$\mathbf{Y}_{\text{lin}} = \mathbf{W}_s\mathbf{X}_s + \mathbf{W}_t\mathbf{X}_t \in \mathbb{R}^{H\times C}, \qquad \mathbf{W}_s, \mathbf{W}_t \in \mathbb{R}^{H\times L}, \tag{3}$$

i.e., a DLinear-style map [11] (randomly initialized and trained jointly — a parameterized pathway, not a fitted DLinear), shared across variates.

Selective-scan correction. The pair $[\mathbf{X}_s; \mathbf{X}_t]$ is embedded per time step and processed channel-independently by a stack of Mamba layers [6] (Fig. 2a). Instantiating the selective recurrence of Section 3.2, the per-step discrete parameters are $\bar{\mathbf{A}}_k = \exp(\boldsymbol{\Delta}_k\mathbf{A})$ and $\bar{\mathbf{B}}_k = \boldsymbol{\Delta}_k\mathbf{B}_k$, and the layer runs the recurrence

$$\mathbf{h}_k = \bar{\mathbf{A}}_k\, \mathbf{h}_{k-1} + \bar{\mathbf{B}}_k\, u_k, \qquad y_k = \mathbf{C}_k^\top \mathbf{h}_k + D\, u_k, \tag{4}$$

scanned causally over the L steps and applied independently to each of the d_{in} feature dimensions: per dimension, the state is $\mathbf{h}_k \in \mathbb{R}^N$, $\mathbf{A} \in \mathbb{R}^N$ is diagonal, the input u_k is scalar, and $(\boldsymbol{\Delta}_k \in \mathbb{R},\ \mathbf{B}_k, \mathbf{C}_k \in \mathbb{R}^N)$ are produced from u by learned linear maps — the input-dependent selectivity of [6]. The last state, summed with a whole-window linear embedding $\mathbf{W}_e\tilde{\mathbf{x}}^{(c)}$ (one per variate, as in [7]), forms the branch feature $\mathbf{f}_{\text{time}} \in \mathbb{R}^{C\times d}$. A head ϕ_{time} maps features to a horizon correction:

$$\mathbf{Y}_{\text{time}} = \mathbf{Y}_{\text{lin}} + \phi_{\text{time}}(\mathbf{f}_{\text{time}}), \qquad \phi_{\text{time}} \equiv \mathbf{0} \text{ at init.} \tag{5}$$

Because ϕ_{time} is zero-initialized, the branch's output at initialization equals its (randomly initialized) DLinear-style backbone; the SSM learns only what the linear map cannot express.

4.2. Frequency Branch: Spectral Filtering and a Dormant Spectral Pathway

The frequency branch applies the real FFT along time, $\mathbf{Z} = \mathrm{rFFT}(\tilde{\mathbf{X}}) \in \mathbb{C}^{C\times F}$, $F = \lfloor L/2 \rfloor + 1$, optionally low-passed by zeroing the top fraction ρ of bins.

Spectral encoder. A learnable complex linear filter $\mathbf{W} = \mathbf{W}_r + i\mathbf{W}_i \in \mathbb{C}^{F\times F}$ densely mixes bins, $\mathbf{Z}' = \mathbf{Z}\mathbf{W}^\top$ along the frequency axis (a bidirectional Mamba over the bin sequence is an input-dependent alternative). The (real, imaginary) parts are projected to the branch feature $\mathbf{f}_{\mathrm{freq}} \in \mathbb{R}^{C\times d}$ and a head ϕ_{freq} produces the spectral forecast.

FITS-style spectral pathway (optional, per dataset). A complex linear map $\mathbf{U} \in \mathbb{C}^{F'\times F}$ with $F' = \lfloor (L+H)/2 \rfloor + 1$, applied along the frequency axis ($\mathbf{Z}\mathbf{U}^\top \in \mathbb{C}^{C\times F'}$), maps the (low-passed) window spectrum to that of a length-($L+H$) series, whose inverse transform's last H samples are the branch's linear spectral forecast:

$$\mathbf{Y}_{\mathrm{freq}} = \underbrace{\mathrm{irFFT}_{L+H}(\mathbf{Z}\mathbf{U}^\top)[\,L{:}\,]}_{\text{linear-in-frequency}} + \phi_{\mathrm{freq}}(\mathbf{f}_{\mathrm{freq}}). \tag{6}$$

This pathway uses the frequency interpolation idea of FITS [12], but is not a fitted FITS model. When the spectral backbone is enabled (Table 2: ETTh1, ETTh2, Weather), both $\mathbf{U}$ and the forecast head ϕ_{freq} are *zero-initialized*, so the frequency branch is silent at initialization; we use a zero rather than an identity init because the length-L and length-($L+H$) frequency grids do not align, so no canonical identity init exists, and a random $\mathbf{U}$ would inject noise during the highest-learning-rate epochs. On the remaining datasets the spectral backbone is disabled: the linear-in-frequency term of Eq. (6) is absent and ϕ_{freq} is *randomly* initialized, so there the frequency branch contributes a small, gate-attenuated signal from the first step rather than being silent. Consequently the clean RLinear-class initialization—in which, with the time head, spectral map, and gate all zero or constant at init, the end-to-end map in the normalized space reduces to the linear $g_0 \cdot \mathbf{Y}_{\mathrm{lin}}$ and, composed with RevIN, to the RLinear-class structure [15, 16]—holds exactly only for the backbone-enabled configurations. Elsewhere it holds approximately, up to the $(1-g_0)$-weighted random frequency term, which the near-saturated fusion gate ($g_0 \approx 0.9$) keeps small.

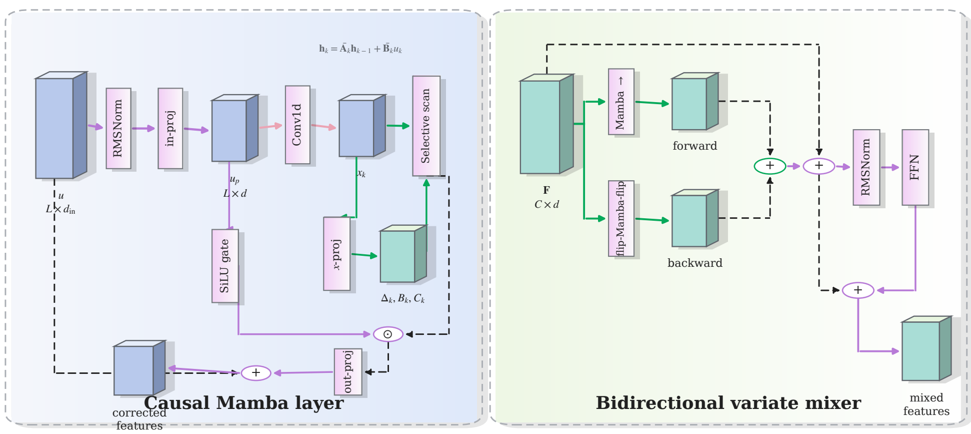

Figure 2: Selective state-space modules used by DD-Mamba. **(a)** The causal time-axis Mamba block normalizes and projects the input, forms a convolutional content stream and a SiLU gate, predicts $(\Delta_k, \mathbf{B}_k, \mathbf{C}_k)$ from the input, and evaluates Eq. (4) with an outer residual connection. **(b)** The variate mixer scans the C channel tokens in both directions, restores the reverse scan to the original order, sums the two streams, and applies residual-normalized position-wise feed-forward refinement. The two panels distinguish temporal causality (ordered time steps) from non-causal cross-channel interaction (channel order has no temporal interpretation).

4.3. Cross-Channel Mixing over Variates

Channel independence is a strong baseline [4, 11] but leaves accuracy on the table when variates are coupled [5, 7, 14]. Inside each branch, before its head, we optionally run M layers of *bidirectional* Mamba over the C per-channel features, followed by a position-wise feed-forward block (Fig. 2b). Because channel order carries no causal direction, each layer sums a forward and a reversed scan in one residual stream. This construction is non-causal, but it is *not* permutation-invariant: an arbitrary reordering of channels can change both scans. We retain each dataset's canonical column order and treat order sensitivity as a limitation rather than attributing set-like symmetry to the mixer.

As a descriptive measure of channel coupling, we compute on the training split

$$\overline{|r|} = \frac{2}{C(C-1)} \sum_{1 \le i < j \le C} \left|\mathrm{corr}(\mathbf{x}^{(i)}, \mathbf{x}^{(j)})\right|. \tag{7}$$

It spans 0.31–0.35 on the ETT family and 0.50–0.91 on Electricity, Traffic, and Solar (Table 1). Raw PCC does not represent lagged, nonlinear, or conditional dependence and can be inflated by common trends; it is therefore used only to organize the analysis. The mixer remains a per-dataset switch whose value is evaluated empirically in Section 5.3.

4.4. Forecast-Level Convex Fusion

Each branch emits a forecast and a feature. The model output is a gated convex combination

$$\mathbf{Y} = g \odot \mathbf{Y}_{\mathrm{time}} + (1-g) \odot \mathbf{Y}_{\mathrm{freq}}, \qquad g = \sigma\big(\psi([\mathbf{f}_{\mathrm{time}}; \mathbf{f}_{\mathrm{freq}}])\big) \in (0,1)^{C\times 1}, \tag{8}$$

with ψ applied row-wise (per variate), followed by RevIN de-normalization. Since $\mathbf{Y}$ lies elementwise between the branch forecasts, both domains retain an explicit prediction path, and the gate value records the convex mixing weight. It is neither an identifiable component contribution nor a causal attribution, because the branches are jointly trained and can compensate for one another. The constraint is structural, not behavioral: g is a sigmoid and can saturate near 0 or 1, so one branch may contribute very little — by design, as the ablations show the useful domain is dataset-dependent. ψ is zero-initialized with its bias set so that training starts gate-open toward the time branch ($g_0 \approx 0.9$; exact value in Section 5.1): the time branch is linear at initialization while the frequency branch is silent, and an even initial mix would halve the initial linear forecast. This is an initialization convention; empirical gate distributions are not available in the present study.

4.5. Training

All branches, gates, and heads are optimized jointly by minimizing the MSE between the forecast and the target. Optimizer, schedule, stopping rule, and dataset-specific settings are reported with the experimental protocol in Section 5.1.3.

5. Experiments

5.1. Setup

5.1.1. Datasets and Protocol

We evaluate on nine standard benchmarks (Table 1) under the unified protocol of [5, 7]: look-back 96, horizons 96/192/336/720. ETT uses the

Table 1: Benchmark datasets, their cross-channel coupling, and where DD-Mamba places its capacity. $\overline{|r|}$: mean absolute pairwise Pearson correlation between channels on the training split (— = raw file unavailable in our build environment).

Dataset	Variates	Steps	Granularity	$\overline{\lvert r\rvert}$	Variate mixer	Params
ETTh1/ETTh2	7	17,420	1 hour	0.31/0.35	off	0.25M
ETTm1/ETTm2	7	69,680	15 min	0.31/0.35	off	0.24–0.36M
Weather	21	52,696	10 min	—	M=2, d=256	5.8M
Solar-Energy	137	52,560	10 min	0.91	M=1, d=128	0.96M
Electricity	321	26,304	1 hour	0.50	M=2, d=512	19.7M
Traffic	862	17,544	1 hour	0.58	M=2, d=512	19.7M
Exchange	8	7,588	1 day	0.48	off	0.09M

canonical 12/4/4-month border split, the others chronological 0.7/0.1/0.2 splits; variates are standardized by training-set statistics; metrics are MSE and MAE.

5.1.2. Baselines

We compare linear, attention-based, and state-space forecasters. The primary comparison uses one evaluation pipeline (Table 4): DLinear [11], NLinear [11], RLinear [15], PatchTST [4], iTransformer [5], S-Mamba [7], and ms-Mamba [8] are re-implemented in our codebase and trained through the same data pipeline, splits, schedule, and evaluation as DD-Mamba, over five matched run indices. These controls reduce split and evaluation confounds, but do not equalize model-specific tuning budgets. The reruns cover six datasets (ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity); seed-matched differences are analyzed as described below.

Published results are not mixed with this ranking because their preprocessing, optimization, and seeds differ. Crossformer [14] is a particularly relevant spatial–temporal baseline, but our current adapter omits its hierarchical encoder–decoder merging and is therefore not presented as a faithful Crossformer result. Likewise, the S-Mamba and ms-Mamba entries are repo-native reproductions rather than outputs from the authors' code; in particular, our ms-Mamba adapter realizes the multi-scale idea by *temporally subsampling* the look-back at rates $\{1, 2, 4\}$ and summing per-scale Mamba encoders, rather than the published design's parallel Mamba blocks with distinct per-scale state-space sampling-rate (Δ) parameters, so it is a multi-scale proxy rather than a faithful ms-Mamba. The accuracy claim is consequently restricted to the seven implemented baselines and six datasets in Table 4; no superiority claim is made against Crossformer, TF4TF, or the three datasets without in-pipeline comparisons.

5.1.3. Implementation and Reproducibility.

All experiments use PyTorch 2.11 on a single NVIDIA RTX 5090. Optimization is AdamW for at most 10 epochs, with the learning rate halved after each epoch, gradient clipping at 5.0, and best-checkpoint early stopping on validation loss. The decomposition kernel is 25; Mamba blocks use expansion 2 and convolution width 4. The search space, selected configurations, and remaining implementation details are reported in Table 2 and the reproducibility package. This schedule was held fixed across runs but was not compared with longer-budget or warm-up alternatives, so schedule sensitivity remains a potential confound. DD-Mamba's per-horizon results (Table 3) are means over *three seeds* (2024–2026); the maximum per-entry std across the 36 dataset–horizon entries is 0.005 MSE, typically 0.001–0.003.

The unified in-pipeline comparison (Table 4) uses *five* matched run indices (2024–2028) for every model, and DD-Mamba's five-seed averages reproduce its three-seed numbers to within 0.002 MSE. The table records final empirical configurations, not a transferable capacity-selection rule. No common pre-registered search space or equalized model-specific tuning budget was used. Consequently, the comparison should be read as a standardized-pipeline evaluation rather than a fully budget-matched benchmark.

5.1.4. Statistical Methodology

We compare at two levels. *Within our framework* (ablations), every variant shares seeds with the reference, so we form the *paired* per-seed difference and report its mean effect, a two-sided t-based p-value, and a 95% CI (n=3, t_{crit}=4.303). Because we run many such tests, raw p-values overstate significance, so we additionally control the false discovery rate with the Benjamini–Hochberg (BH) procedure *within each family* — the 29 component-placement comparisons and the 54 branch-removal comparisons — and report the resulting q-values (`scripts/compute_stats_correction.py`). An effect is called statistically supported only if it survives BH at q<0.05 and its configuration was not selected using the same test set; all other effects are reported as exploratory. For the ablations we flag n=3 as a substantial limitation of power; effect sizes with CIs are reported throughout so that readers can judge magnitude independently of the significance verdict. *Unified in-pipeline comparison* (Table 4): we test each DD-Mamba-vs-baseline difference on average-over-horizons MSE using the five matched run indices and a paired two-sided t-test (t_{crit}=2.776), followed by BH correction across the comparison family (`scripts/analyze_unified_baselines.py`). The paired interpretation relies on matching run index and data order; with only five runs, effect magnitudes and uncertainty are considered alongside the corrected tests.

Table 2: Per-dataset hyperparameters. d: model width; time enc.: time-branch encoder (Mamba layers in parentheses); M: variate-mixer layers; N: SSM state size; ρ: spectral low-pass fraction (Section 4.2); spec. map: the *optional* FITS-style pathway of Eq. (6) — when "no", the frequency branch remains fully active (rFFT, spectral encoder, forecast head) but carries no explicit linear pathway; IN: instance normalization (RevIN); do.: dropout; B: batch size; wd: weight decay. †Solar adopts IN = no after inspection of a test-set ablation (Section 5.3); all results under this setting are labeled exploratory and excluded from confirmatory claims. The widths listed here are the configurations that produced the reported accuracy, and the repository's released configs match them. The variate mixer is instantiated independently in both branches (`mixer_placement=both`) except on Electricity and Traffic, where weight-tying it across branches was measured accuracy-neutral and is used to halve the mixer cost (Section 6). Parameter-reduced variants of these models, and the accuracy they cost, are reported separately in Table 10; they are not the configurations behind the results tables. §Electricity's reported numbers were produced with the spectral pathway disabled and therefore with a randomly initialized frequency head; Section 6 argues this accounts for part of its short-horizon deficit. Enabling the pathway was tested and made the dataset average worse ($0.169 \rightarrow 0.175$, almost entirely at H=720), so it remains disabled.

Dataset	d	time enc.	M	N	ρ	spec. map	IN	lr	B	do./wd
ETTh1	128	Mamba (1)	0	16	0.4	yes	yes	$5{\cdot}10^{-4}$	32	$0.2/10^{-4}$
ETTh2	128	Mamba (1)	0	16	0.3	yes	yes	$5{\cdot}10^{-4}$	32	$0.3/5{\cdot}10^{-4}$
ETTm1	128	Mamba (2)	0	16	0	no	yes	10^{-4}	32	$0.1/10^{-4}$
ETTm2	128	Mamba (1)	0	16	0.2	no	yes	10^{-4}	32	$0.2/2{\cdot}10^{-4}$
Weather	256	Mamba (2)	2	16	0	yes	yes	$2{\cdot}10^{-4}$	32	$0.1/10^{-4}$
Solar	128	Mamba (2)	1	16	0	no	no†	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Electricity	512	MLP	2	32	0	no§	yes	$5{\cdot}10^{-4}$	16	$0.1/10^{-4}$
Traffic	512	MLP	2	32	0	no	yes	10^{-3}	16	$0.1/10^{-4}$
Exchange	64	Mamba (1)	0	8	0.5	no	yes	10^{-4}	32	$0.3/5{\cdot}10^{-4}$

Table 3: DD-Mamba per-horizon results, look-back L=96, from the most recent run of the released configurations. Solar uses a test-informed normalization setting and is reported only as an exploratory diagnostic result. Values are point estimates from this run; the seed-level dispersion quoted elsewhere in the paper comes from the three-seed study of the same configurations, whose per-horizon means agree with these to within ±0.003 MSE on every dataset. Electricity and Weather were re-run at the capacity Table 2 lists after the reduction study; Table 10 gives the separately measured cost of their parameter-reduced variants.

Dataset		H=96	H=192	H=336	H=720	Avg
ETTh1	MSE	0.381	0.429	0.472	0.472	0.439
	MAE	0.397	0.423	0.446	0.468	0.434
ETTh2	MSE	0.292	0.368	0.415	0.423	0.374
	MAE	0.345	0.392	0.427	0.442	0.402
ETTm1	MSE	0.321	0.366	0.396	0.461	0.386
	MAE	0.359	0.385	0.405	0.443	0.398
ETTm2	MSE	0.177	0.241	0.302	0.398	0.280
	MAE	0.260	0.302	0.341	0.397	0.325
Weather	MSE	0.162	0.212	0.267	0.349	0.247
	MAE	0.207	0.254	0.293	0.347	0.275
Electricity	MSE	0.141	0.159	0.174	0.200	0.169
	MAE	0.236	0.253	0.270	0.296	0.264
Traffic	MSE	0.401	0.423	0.441	0.481	0.437
	MAE	0.268	0.276	0.284	0.303	0.283
Exchange	MSE	0.086	0.179	0.329	0.858	0.363
	MAE	0.206	0.301	0.415	0.699	0.405
Solar	MSE	0.181	0.189	0.207	0.206	0.196
	MAE	0.241	0.256	0.269	0.269	0.259

Table 4: *Unified in-pipeline* comparison (average MSE/MAE over $H \in \{96, 192, 336, 720\}$, look-back L=96; five matched run indices; lower is better). Every baseline is re-implemented and evaluated with the same splits and protocol as DD-Mamba; model-specific tuning budgets were not equalized. Following the presentation of S-Mamba [7] and ms-Mamba [8], the **best** result per dataset and metric is bold and the second best underlined. * DD-Mamba beats the best competing baseline at Benjamini–Hochberg q<0.05 (paired t-test, n=5; `analyze_unified_baselines.py`); those papers report point estimates without significance testing, so the starred cells are the subset of wins this protocol can actually support. DD-Mamba has the lowest MSE on five of six datasets and the corrected test supports the margin on four; Weather is a statistical tie (q=0.23) and PatchTST is better on ETTm1.

Models	**DD-Mamba**		PatchTST		iTransformer		S-Mamba		ms-Mamba		RLinear		NLinear		DLinear	
Metric	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
ETTh1	**0.439***	**0.434**	0.456	0.449	0.444	0.442	0.463	0.458	0.459	0.451	0.457	0.443	0.464	0.450	0.462	0.456
ETTh2	**0.376***	**0.401**	0.385	0.408	0.383	0.407	0.394	0.410	0.409	0.420	0.391	0.411	0.390	0.412	0.545	0.509
ETTm1	0.387	0.398	**0.381**	**0.396**	0.407	0.409	0.397	0.406	0.393	0.403	0.424	0.413	0.429	0.419	0.410	0.412
ETTm2	**0.280***	**0.325**	0.284	0.331	0.292	0.335	0.297	0.335	0.288	0.333	0.293	0.336	0.294	0.337	0.359	0.407
Weather	**0.248**	0.276	0.249	**0.275**	0.265	0.285	0.251	0.278	0.257	0.281	0.273	0.292	0.279	0.296	0.267	0.318
Electricity	**0.168***	**0.263**	0.189	0.276	0.195	0.282	0.183	0.275	0.187	0.277	0.217	0.295	0.220	0.299	0.211	0.299

Table 5: PEMS results against the *published* numbers of S-Mamba [7] and ms-Mamba [8], look-back L=96, horizons $T \in \{12, 24, 48, 96\}$ (MSE). **Best** bold, second best underlined. **These are not same-pipeline reruns.** Unlike Table 4, the competitors' figures are transcribed from their papers and were produced by their own code, preprocessing and seeds, so this table is a positioning reference rather than a controlled comparison, and no significance test is applicable to it. PEMS is a confirmatory family for us: these datasets were not used to develop any component of DD-Mamba.

Dataset	Model	T=12	T=24	T=48	T=96	Avg
PEMS03	DD-Mamba	<u>0.065</u>	**0.085**	**0.128**	**0.187**	**0.116**
	S-Mamba	**0.065**	<u>0.087</u>	<u>0.133</u>	<u>0.201</u>	<u>0.122</u>
	ms-Mamba	0.066	<u>0.087</u>	<u>0.133</u>	<u>0.201</u>	0.122
PEMS04	DD-Mamba	0.075	0.093	0.123	0.171	0.116
	S-Mamba	<u>0.076</u>	<u>0.084</u>	<u>0.115</u>	<u>0.137</u>	<u>0.103</u>
	ms-Mamba	**0.072**	**0.083**	**0.099**	**0.121**	**0.094**
PEMS07	DD-Mamba	0.059	0.079	0.105	0.139	0.095
	S-Mamba	<u>0.063</u>	<u>0.081</u>	<u>0.093</u>	<u>0.117</u>	<u>0.089</u>
	ms-Mamba	**0.060**	**0.075**	**0.091**	**0.109**	**0.084**
PEMS08	DD-Mamba	**0.074**	**0.099**	**0.146**	**0.234**	**0.138**
	S-Mamba	<u>0.076</u>	0.104	0.167	0.245	0.148
	ms-Mamba	<u>0.073</u>	<u>0.098</u>	<u>0.154</u>	<u>0.236</u>	<u>0.140</u>

On the four PEMS datasets (Table 5) DD-Mamba is ahead of both published Mamba forecasters on PEMS03 (0.116 vs. 0.122/0.122) and PEMS08 (0.138 vs. 0.148/0.140), and behind both on PEMS04 and PEMS07, where ms-Mamba's multi-scale sampling rates are strongest. Because the competing figures come from their own pipelines, we read this as *positioning* rather than as evidence of superiority: the split is two datasets each way, and the protocol difference is large enough that a 0.005-scale margin should not be over-interpreted.

Table 4 is the paper's primary accuracy evidence because it uses one data and evaluation protocol. Under this five-run protocol DD-Mamba has the lowest average MSE on five of the six re-run datasets, and the paired, BH-corrected test supports the margin on **four**: Electricity (0.168 vs. S-Mamba's 0.183; $q \ll 0.01$), ETTh2 (0.376 vs. iTransformer's 0.383; q=0.001), ETTm2 (0.280 vs. PatchTST's 0.284; q=0.015), and ETTh1 (0.439 vs. iTransformer's 0.444; q=0.021). Weather is a statistical *tie* — DD-Mamba is numerically best (0.248 vs. PatchTST's 0.249) but the difference does not survive correction (q=0.23). On ETTm1, PatchTST is lower (0.381 vs. 0.387; q<0.001), consistent with the branch matrix, where ETTm1 is carried by the time branch and its patch-attention competitor is strong. DD-Mamba's size is dataset-dependent: on the low-channel ETT and Exchange datasets it uses only $\approx$ 0.1–0.25M parameters (an order of magnitude fewer than the variate-attention baselines), whereas on high-channel data the variate mixer dominates the count and the model is larger (Table 11); that mixer cost is itself reducible, as the complexity analysis below notes. These results do not establish a ranking on Solar, Traffic, or Exchange, where comparable in-pipeline baselines are unavailable.

5.3. Component Ablations

Our ablation harness flips exactly one switch of a dataset's full configuration, holding split, schedule, seeds, and every other hyperparameter fixed. The design follows S-Mamba [7], whose component study groups variants into a *reference* row, a *Replace* block that swaps a component for an alternative of the same role, and a *w/o* block that removes it outright, reporting each block's effect on both metrics; ms-Mamba [8] adds the complementary idea of sweeping a design *choice* rather than only deleting parts. We adopt both. Table 6 lays out the resulting grid over the three blocks of DD-Mamba: the cross-variate mixer (S-Mamba's *variate correlation* block), the time-axis encoder (their *temporal dependency* block), and the frequency branch, which has no S-Mamba counterpart.

Two consequences follow from taking their design seriously. First, S-Mamba benchmarks its bidirectional Mamba mixer against multi-head attention and against a unidirectional scan; we implement both alternatives (`mixer_kind`) so that comparison runs inside our pipeline rather than being inherited across papers. Second, because several switches are already set to one of their two values by a given dataset — PEMS ships an MLP time encoder and a weight-tied mixer, ETT ships Mamba and untied — the grid is resolved per configuration, so no row silently duplicates the reference and inflates the correction family.

A methodological difference is worth stating explicitly, because it changes what the table can support. S-Mamba and ms-Mamba both report single-run deltas and read the sign directly. With twelve variants, roughly one would

Table 6: Ablation design, following S-Mamba's grouping into a reference row, *Replace* rows (component swapped for an alternative of the same role) and *w/o* rows (component removed). VC: cross-variate mixer; TD: time-axis encoder; FD: frequency branch. "base" is whatever the dataset's released configuration specifies, so the grid resolves to 12 runs on the high-channel datasets and 7 where no mixer is configured.

Design	Variant	VC	TD	FD
reference	full	bi-Mamba	base	linear filter
Replace	uni-directional mixer	uni-Mamba	base	linear filter
	attention mixer	Attention	base	linear filter
	mixer placement	tied ↔ untied	base	linear filter
	mixer in time branch only	time only	base	linear filter
	time encoder	bi-Mamba	Mamba ↔ MLP	linear filter
	spectral encoder	bi-Mamba	base	bi-Mamba
	fusion rule	bi-Mamba	base	linear filter (averaged)
w/o	no cross-variate mixing	w/o	base	linear filter
	no frequency branch	bi-Mamba	base	w/o
	no time branch	bi-Mamba	w/o	linear filter
	no instance normalization	bi-Mamba	base	linear filter

clear an uncorrected 0.05 threshold by chance, so we run multiple seeds, report paired confidence intervals, and apply Benjamini–Hochberg across the family. That is why our component conclusions are weaker than theirs on the same architecture family: not because the effects are smaller, but because they are being asked to survive a correction. We report its complete output on three datasets under their *final released configurations*, three seeds each at H=96: ETTh1 and Solar (Table 7) and Weather (Table 9), with the paired methodology of Section 5.1.3. Two consistency checks pass: the Solar "full" row reproduces the main-table Solar H=96 entry (0.180$\pm$.004), and its w/o-instance-norm variant coincides with the full (RevIN-off) configuration, so

Table 7: Component ablations on ETTh1 and Solar-Energy (H=96, three seeds, final configurations): paired per-seed ΔMSE to the full model with t-based 95% CIs; *CI excludes zero. The spectral map is enabled on ETTh1 (hence *removed*) and disabled on Solar (hence *added*); the variate mixer is off on ETTh1 (*added*) and on for Solar (*removed*).

Variant	ETTh1: Δ [95% CI]	Solar: Δ [95% CI]
full (reference MSE)	0.3802±.001	0.1802±.004
time branch only (freq removed)	+0.0034 [−0.009, +0.016]	+0.0114 [+0.001, +0.021]*
freq branch only (time removed)	−0.0007 [−0.004, +0.003]	+0.0051 [−0.013, +0.023]
sum fusion	−0.0026 [−0.007, +0.002]	+0.0066 [−0.005, +0.018]
w/o instance norm	+0.0061 [−0.001, +0.013]	—
w/o linear backbone	−0.0031 [−0.009, +0.003]	+0.0081 [−0.019, +0.035]
random init (vs. zero-init)	−0.0012 [−0.005, +0.002]	−0.0046 [−0.022, +0.013]
spectral map removed / added	+0.0042 [−0.018, +0.026]	+0.0088 [−0.002, +0.019]
variate mixer added / removed	+0.0077 [+0.004, +0.012]*	+0.0119 [−0.010, +0.034]
Mamba→MLP over time	−0.0022 [−0.007, +0.003]	+0.0089 [−0.002, +0.020]
freq Mamba encoder	−0.0024 [−0.011, +0.006]	−0.0117 [−0.044, +0.020]

that cell is omitted.

Component-placement effects are directional but exploratory.. A two-sided pattern spans datasets at the raw level: adding the variate mixer on ETTh1 *hurts* (by +0.0077 MSE, raw p=0.014) while removing it on Weather also hurts (by +0.0104, raw p=0.022); normalization shows the same shape (removing it hurts Weather +0.0161, raw p=0.023, while disabling it on Solar improved the re-measured main results from 0.228 to 0.199). We emphasize, however, that across the 29 component-placement comparisons *none* survives Benjamini–Hochberg correction (the five raw-significant effects have $q \geq 0.11$): at three seeds these are *exploratory, directional* findings, not confirmed effects, and we report them as such. The consistency of the *direction* (component value is dataset-dependent, in both directions) is what motivates per-dataset switches; supporting any individual placement effect requires more runs and validation-only configuration selection. Zero initialization is a *null* result — no significant difference on either dataset (−0.0012 ETTh1, −0.0046 Solar; CIs include zero) — so we make no accuracy claim for it, only the interpretable linear start (Section 4.1). The DLinear backbone and the learned gate versus averaging are within CIs everywhere; on Weather (Table 9) three components have unadjusted paired CIs excluding zero (normalization, the variate mixer, and the temporal Mamba, +0.0027, positive on all seeds). The recurring pattern — component value is dataset-dependent, in *both* directions — is the argument for per-dataset switches over universal defaults, and it does not depend on any single dataset.

Branch matrix: where each domain matters.. Because a single Solar cell cannot carry a dual-domain conclusion, we ran the full branch-ablation matrix — full / time-only / freq-only under paired seeds on all nine datasets at H=96 and across all four horizons on the six ETT/Weather/Solar datasets, 27 cells in total (Table 8). Every full-model mean reproduces the corresponding main-table entry, and the Solar H=96 cell replicates the earlier ablation (+0.0126 vs. +0.0114, overlapping CIs). The 27 cells yield 54 branch-removal tests; we report both raw significance (CI excludes zero, marked * in Table 8) and BH-corrected q-values, visualized in Fig. 3. A corrected result is treated as statistically supported only when its configuration was not chosen using the same test set.

Frequency branch. At the raw level, removing it is significant in 5 cells, concentrated at the shortest horizon: Solar (+0.0126), ETTm1 (+0.0073), Electricity (+0.0019), and Exchange (+0.0015) at H=96, plus ETTm2 at

H=336. After correction, however, only **Solar at** H=96 survives (q=0.048); the other four fall to $q \geq 0.10$. So the frequency-side result is narrow and, because Solar's normalization setting was selected with test feedback, remains *exploratory.*

Time branch. Removing it is significant in 9 cells raw and **8 survive BH** (q<0.05): ETTm1 at *every* horizon (+0.017–+0.021, the most persistent effect we measure), ETTm2 at H=96 and H=336, and Electricity and Exchange at H=96. The time pathway is therefore the only branch with repeated corrected evidence, especially on ETTm1. On ETTh1, ETTh2, and Weather no branch effect is detected at any horizon; this is absence of detected evidence, not evidence that the branches are interchangeable.

The matrix does not establish a consistent dual-domain advantage. Instead, it shows repeated sensitivity to the temporal branch and only a test-informed frequency-side effect. A branch with a small fitted gate weight remains computationally active, so it should not be described as cost-free redundancy.

Table 8: Branch-ablation matrix (27 cells): paired per-seed ΔMSE versus the full model when one branch is removed (three seeds; * marks *raw* t-based 95% CI excluding zero; per-cell CIs ship with the results). After Benjamini–Hochberg correction over the 54 tests, 9 survive at $q<0.05$ — eight on the time side (ETTm1 all horizons; ETTm2@96/336; Electricity@96; Exchange@96) and, on the frequency side, *only* Solar@96 (q=0.048); the remaining starred cells are exploratory. The Solar result is also exploratory because its normalization setting was selected with test feedback. Electricity, Traffic, and Exchange were run at H=96 only (—). Full-model means match Table 3.

	freq removed: ΔMSE at H				time removed: ΔMSE at H			
Dataset	96	192	336	720	96	192	336	720
ETTh1	+.0034	+.0005	−.0047	+.0096	−.0007	−.0001	−.0003	+.0176
ETTh2	+.0006	+.0078	+.0054	+.0093	−.0031	−.0017	−.0017	+.0026
ETTm1	+.0073*	+.0002	+.0019	−.0005	+.0196*	+.0168*	+.0203*	+.0210*
ETTm2	+.0020	+.0014	+.0030*	+.0001	+.0060*	+.0060	+.0059*	+.0059
Weather	+.0008	+.0004	−.0002	−.0009	+.0006	+.0011	+.0021	+.0027
Solar	+.0126*	−.0038	+.0006	−.0032	+.0052	+.0075	+.0040*	+.0027
Electricity	+.0019*	—	—	—	+.0017*	—	—	—
Traffic	−.0015	—	—	—	+.0013	—	—	—
Exchange	+.0015*	—	—	—	+.0058*	—	—	—

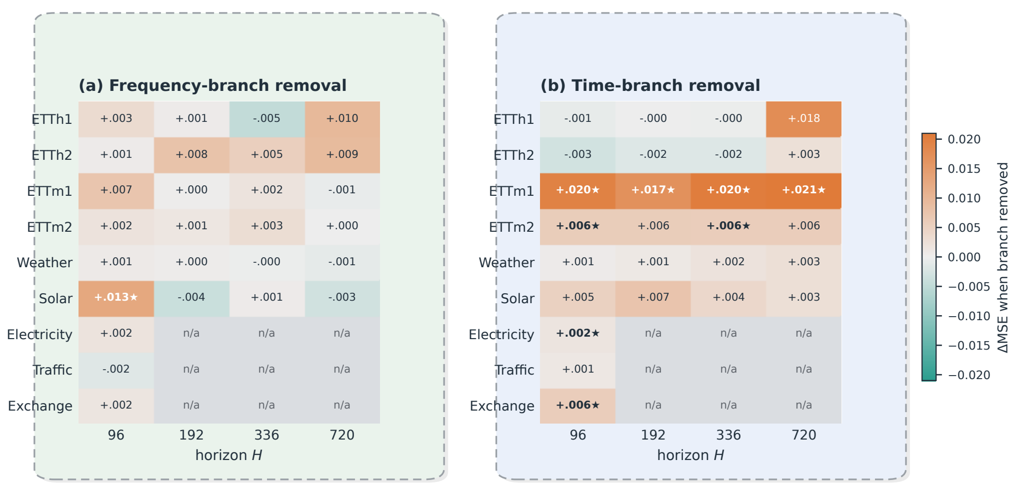

Figure 3: Branch-removal effect (ΔMSE) across datasets and horizons, visualizing Table 8: **(a)** removing the frequency branch and **(b)** removing the time branch. Warm cells mark configurations where removal *hurts* (the branch is valuable), cool cells where removal *helps* (the branch is redundant), and gray where there is no effect; a ★ marks effects surviving Benjamini–Hochberg correction (q < 0.05). Only Solar@96 survives on the frequency side, whereas the time branch is confirmed across the ETTm datasets and at H=96 on

Table 9: Component ablation on Weather (H=96; MSE mean ± std over three seeds). Δ is the *paired* per-seed difference to the full model with a t-based 95% confidence interval (n=3); * CI excludes zero.

Variant	MSE	paired Δ [95% CI]	Component isolated
full	0.1620±.0003	—	reference
w/o instance norm	0.1781±.0035	+0.0161 [+0.0053, +0.0269]*	RevIN
w/o channel mixer	0.1724±.0021	+0.0104 [+0.0037, +0.0172]*	cross-variate Mamba mixing
Mamba→MLP over time	0.1647±.0005	+0.0027 [+0.0020, +0.0034]*	selective scan (time)
w/o linear backbone	0.1631±.0006	+0.0011 [−0.0014, +0.0036]	DLinear anchor (time)
time branch only	0.1628±.0011	+0.0007 [−0.0020, +0.0035]	frequency branch removed
w/o spectral map	0.1627±.0016	+0.0007 [−0.0041, +0.0055]	FITS-style pathway
freq branch only	0.1626±.0014	+0.0006 [−0.0040, +0.0052]	time branch removed
freq Mamba encoder	0.1626±.0016	+0.0006 [−0.0044, +0.0055]	spectral encoder type
sum fusion	0.1622±.0007	+0.0001 [−0.0020, +0.0023]	learned gate → average

Negative results that shaped the design.. Deepening the traffic mixer from two to four layers (19.7 → 37.9M) gave no gain, so the release keeps two; enabling the mixer on ETT (8.4K training windows) tripled parameters and was associated with premature early stopping and worse held-out error, so the released ETT configurations disable it.

5.4. Analysis

Where branch removals matter.. The matrix (Table 8) shows that the temporal path has the more repeatable effect: eight removals survive correction, including every ETTm1 horizon. The sole corrected frequency-side cell is Solar at H=96, where the 10-minute daily cycle spans $144 > L = 96$ samples. This periodicity offers a plausible hypothesis for the short-horizon effect, but the mechanism is not identified by the ablation and the result remains exploratory because the Solar configuration used test feedback. On ETTh1, ETTh2, Weather, and Traffic, the study does not detect a branch-removal effect under the tested settings. Such null results should not be interpreted as equivalence, especially at three seeds.

Cross-channel correlation as a descriptive covariate.. The mixer-off ETT family has $\overline{|r|} = 0.31$–0.35, whereas Electricity, Traffic, and Solar, whose released configurations include a mixer, have values of 0.50–0.91. This alignment motivates a correlation-to-capacity hypothesis, but it does not validate one: Weather lacks a measured PCC, Exchange has $\overline{|r|} = 0.48$ with the mixer disabled, and the switch settings were not generated by a preregistered threshold. Moreover, raw PCC can reflect common trends rather than exploitable conditional dependence. Accordingly, PCC is reported as a dataset descriptor. A predictive selector would require lag-aware and nonlinear dependence measures, validation-only threshold selection, and evaluation on held-out datasets.

The error profile is horizon-dependent, and initialization explains part of it.. On Electricity the margin against the strongest published competitor varies monotonically with the horizon: relative to the S-Mamba average, DD-Mamba is +1.4% worse at H=96, level at H=192, then 1.1% and 2.0% better at H=336 and H=720. Weather shows no such deficit and is ahead at every horizon. The two datasets differ in a way that predicts exactly this: Weather enables the FITS spectral pathway of Eq. (6) and Electricity does not, and as noted in Section 4.2 the frequency forecast head is zero-initialized *only* when that pathway is present. Electricity therefore begins training with a randomly initialized frequency head whose output enters the forecast at the gate's initial weight $1 - g_0 \approx 0.1$.

We measured that contribution directly. In the normalized space the untrained frequency branch of the Electricity configuration emits a signal of RMS ≈ 0.58, so the injected perturbation is ≈ 0.058 RMS, and it is essentially *constant across horizons* (0.058, 0.057, 0.058, 0.057 at H=96, 192, 336, 720); with the spectral pathway enabled it is exactly zero. A horizon-independent perturbation consumes a larger share of a smaller error budget, and the achievable error here grows from 0.141 at H=96 to 0.200 at H=720 — so the same initialization noise is worth 2.4% of the short-horizon budget but only 1.7% of the long-horizon one, in the same direction and rough proportion as the observed gap.

The obvious remedy — enabling the spectral pathway on Electricity, as Weather does — was tried and *failed*, in an informative way. It left H=336 unchanged and cost +0.002 and +0.001 at H=96 and H=192, but degraded H=720 by +0.024 (0.200 $\rightarrow$ 0.224), moving the dataset average from 0.169 to 0.175. The reason is that the pathway of Eq. (6) interpolates the length-L spectrum onto a length-$(L+H)$ grid, so the map's fan-out grows with the horizon: for L=96 it is 49 $\rightarrow$ 97 bins at H=96 but 49 $\rightarrow$ 409 at H=720, an 8.3$\times$ extrapolation from the same 49 measured coefficients. The damage tracks that ratio monotonically. Our Electricity configuration also applies no spectral low-pass (ρ=0), whereas the two ETT datasets that use this pathway pair it with $\rho \in \{0.3, 0.4\}$; the cutoff is the regularizer that makes linear-in-frequency forecasting stable, and without it the map extrapolates the noisiest high-frequency bins furthest.

This separates two things the "enable FITS" intervention conflates. What the measurement implicates is the *random initialization of the frequency head*, not the absence of a spectral map — and the head can be silenced on its own. We therefore expose the head's initialization as an independent switch: it can be zero-initialized whether or not a backbone is present, at no parameter cost and without introducing any horizon-dependent extrapolation. That is the intervention the evidence supports, and it is the one we would test next; the FITS result above stands as a recorded negative.

We present the initialization account as the leading explanation rather than a demonstrated cause. The horizon trend is measured over four points, and any quantity monotone in H would correlate with it, so the load-bearing evidence is the direct measurement plus the Weather/Electricity contrast, not the correlation. Two further mechanisms point the same way and are not separated by these data. First, the architecture is linear-dominated by construction (zero-initialized corrections, $g_0 \approx 0.9$, and a schedule that halves the learning rate each epoch over ten epochs), and a near-linear predictor is relatively stronger at long horizons, where forecasts approach seasonal climatology, than at short ones, where recoverable local dynamics reward nonlinear capacity. Second, the trend/seasonal split uses a moving average of width 25, which at hourly sampling is roughly one day and smooths away precisely the sub-daily detail that a 96-step forecast could still exploit. Separating them requires a head-initialization ablation, a training-budget sweep, and a kernel-width sweep.

Complexity accounting.. Table 11 reports model-only, hardware-independent estimates rather than comparative runtime evidence. Parameter count is governed by model *width*, not channel count (electricity and traffic share 19.7M despite a 2.7$\times$ gap in variates), so model size is decoupled from data dimensionality; compute and activation memory instead scale with the channel count through the variate mixer (traffic 60 GMACs vs. electricity's 22 at equal width, while ETT and exchange sit two orders of magnitude below), keeping the capacity restraint on those datasets inexpensive. The variate mixer accounts for over 90% of the parameters on high-channel data (it scales with the square of the width and is instantiated in both branches), so its cost is directly tunable: halving the width takes the Electricity and Traffic models from 19.7M to 5.3M, and *weight-tying* one mixer across the two branches — a configuration switch in our implementation — halves it again to 2.9M, bringing DD-Mamba into the size range of the variate-attention baselines. On Traffic this reduction is empirically validated: at H=96 (three seeds) the tied-mixer model matches the untied one within noise (ΔMSE = +0.0007, 95% CI [−0.003, +0.005]), whereas placing the mixer in the time branch *only* is worse in the three-run estimate (+0.0045, unadjusted CI excludes zero). Thus, weight tying reduces the measured parameter cost without a detected Traffic accuracy loss; it is not evidence of zero cost or universal transfer to other datasets.

We subsequently ran the reduction on Electricity and Weather, and it does *not* transfer (Table 10). Both reduced models are worse, and the two cases separate the mechanism. On Electricity, tying the mixer at full width is accuracy-neutral, matching Traffic; it is *halving the width* 512 → 256 that costs +0.0045 average MSE (0.1678 → 0.1722) — enough to move DD-Mamba from better than to worse than the published S-Mamba average of 0.170. Electricity has 321 channels and the strongest cross-channel coupling in the suite ($\overline{|r|}$=0.50), so the mixer capacity that the width controls is doing mea-

Table 10: Measured cost of the parameter-reduced variants. "Shared" ties one variate mixer across the two branches; "d" is the model width. Δ is the change in MSE averaged over $H \in \{96, 192, 336, 720\}$ relative to the shipped configuration (positive is worse). Reductions that were free on Traffic are not free on Electricity or Weather.

Dataset	Configuration	Params	Avg MSE	Δ	Verdict
Traffic	d=256, shared	2.9M	—	+0.0007	neutral (CI incl. 0)
Electricity	d=512, shared	10.6M	0.1678	—	shipped
	d=256, shared	2.8M	0.1722	+0.0045	loses to S-Mamba
Weather	d=256, both	5.8M	0.2485	—	shipped
	d=256, shared	3.5M	0.2497	+0.0013	erases the margin

surable work. On Weather, by contrast, the width is unchanged and merely *tying* the two mixers costs +0.0013 average MSE, concentrated at the longest horizon (0.352→0.356 at H=720): with only 21 channels the mixer is small either way, but the two branches evidently require *different* cross-channel mixing rather than a shared one.

Two conclusions follow. First, sharing a mixer and shrinking a mixer are not interchangeable economies: the former was free on Traffic and Electricity and the latter was not. Second, the earlier Traffic-only result should not have been read as license to reduce elsewhere — the transfer failed on the first two datasets we tested it on. We therefore report the full-capacity configurations in Tables 3 and 4 and treat Table 10 as the measured accuracy–size trade-off curve rather than as a free saving. Table 11 reports the widths that produced the accuracy numbers above. Wall-clock throughput and matched-baseline runtime remain unmeasured.

Table 11: Model-only complexity profile of DD-Mamba per dataset (look-back 96, horizon 96, batch 1). GMACs: multiply–accumulates per forecast (`Linear` + `Conv1d` + the selective-scan recurrence); act. mem.: forward activation footprint under a 32-bit representation. All three columns are hardware-independent estimates; no wall-clock or matched-baseline runtime is claimed.

Dataset	Variates	Params (M)	GMACs	Act. mem. (MB)
Exchange	8	0.09	0.02	2.3
ETTh1	7	0.25	0.08	4.0
Weather	21	5.77	1.90	46.6
Solar	137	0.96	3.26	150.4
Electricity	321	19.70	22.4	254.3
Traffic	862	19.70	60.2	682.7

6. Discussion

6.1. Data characteristics as validation hypotheses

The placement results suggest three hypotheses for future validation, not a deployment rule. First, stronger dependence between channels may justify testing a variate mixer (Fig. 4), but raw PCC ignores lagged and nonlinear structure. Second, a spectral path may help when the look-back does not resolve dominant periods; the present frequency evidence is based on a test-informed setting and does not establish this relationship. Third, normalization may interact with changing scale and periodic zeros. Each choice must be selected on validation data, frozen before test evaluation, and compared with a fixed configuration.

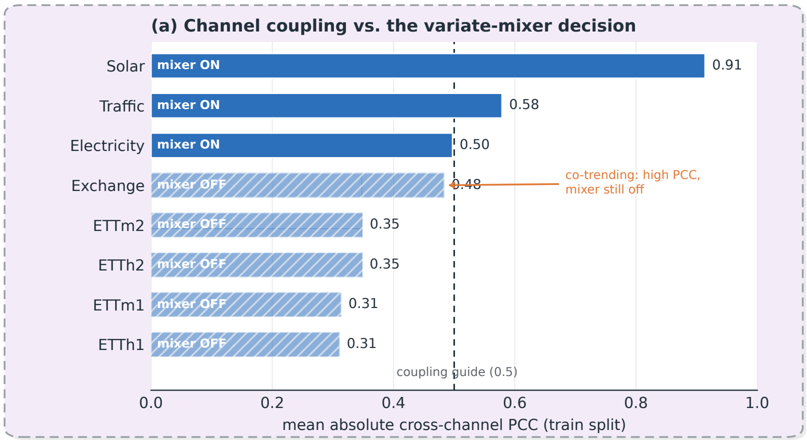

Figure 4: Channel coupling versus the variate-mixer decision. Bars give the mean absolute cross-channel Pearson correlation on the training split; solid bars are datasets where the mixer is enabled, hatched bars where it is disabled. Stronger coupling coincides with enabling the mixer, but Exchange is the instructive exception: its moderate correlation is co-trending rather than exploitable cross-sectional signal, so the mixer stays off. This is why we treat PCC as a screening *hypothesis*, not a selection rule. Weather is omitted (training-split PCC unavailable in this study).

6.2. Dispersion diagnostics: why normalization is dataset-dependent

A training-free analysis explores the two-sided RevIN result (Fig. 5). Standardizing each channel and taking a rolling standard deviation over the dominant period, all three probed datasets carry substantial time-varying scale (coefficient of variation of the rolling std 0.20–0.48); a KPSS test rejects stationarity of the rolling-std series in every sampled channel, and a Ljung-Box test finds it strongly autocorrelated. These tests describe nonstationary scale dynamics but do not prove forecastability or identify a RevIN mechanism. In the measured examples, removing normalization increases

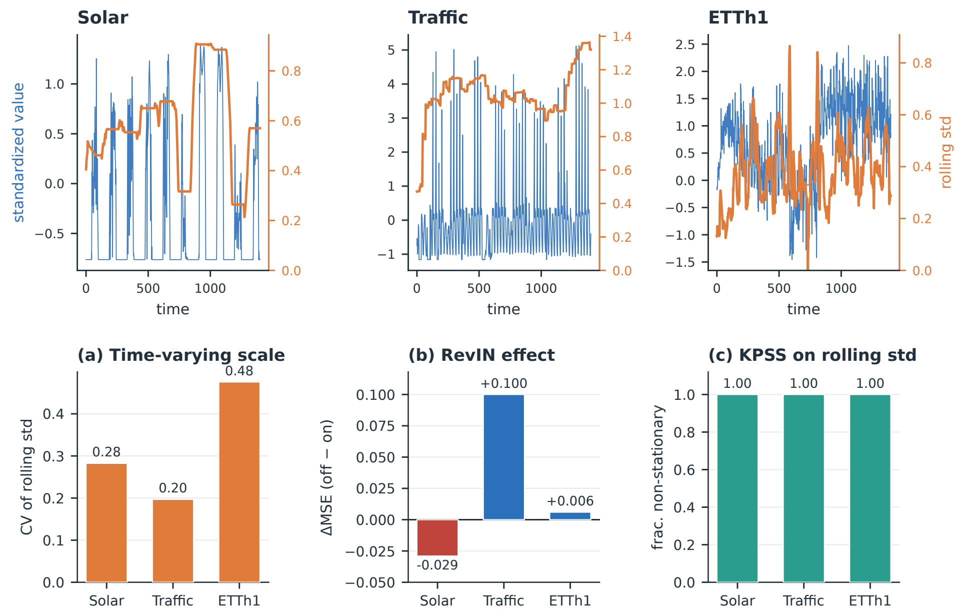

Figure 5: Training-free dispersion diagnostics. *Top:* a representative standardized channel (blue) with its rolling standard deviation (orange) for Solar, Traffic, and ETTh1. **(a)** Coefficient of variation of the rolling std. **(b)** Measured RevIN effect (ΔMSE, off−on): normalizing by the input scale helps Traffic but hurts Solar. **(c)** Fraction of sampled channels whose rolling-std series a KPSS test declares non-stationary. These summaries are descriptive; the Solar effect uses a test-informed configuration.

Traffic MSE by 0.100 and decreases Solar MSE by 0.029. Periodic night-time zeros provide one possible explanation for Solar, but that explanation and the associated test-informed effect require validation on untouched data. Explicit dispersion forecasting in the style of STD [25] is outside the evaluated model.

6.3. Threats to validity

Several threats remain, each with the experiment that would resolve it.

Baseline scope and tuning.. The in-pipeline comparison covers six datasets and seven baselines. It does not include Solar, Traffic, Exchange, Crossformer, or TF4TF, and the S-Mamba/ms-Mamba entries are repo-native implementations rather than outputs from the authors' code. Moreover, one common schedule does not guarantee equal model-specific tuning. The resulting accuracy claim is explicitly restricted to Table 4; broader ranking requires official implementations or verified adapters and comparable search budgets.

The accuracy ceiling is generalization, not optimization.. Our schedule sets $\mathrm{lr} = \mathrm{base} \cdot 0.5^{(e-1)}$, whose summed budget is the geometric series $\sum_e 0.5^{e-1} = 2 \cdot \mathrm{base}$ — *bounded regardless of the epoch count*, with only about four epochs above 10% of the base rate. Because the Electricity training loss is still falling at the final epoch at every horizon, this looked like undertraining, and we tested it directly (`scripts/run_schedule_sweep.py`): the same models were retrained under warmup-plus-cosine over thirty epochs, roughly five times the integrated learning rate.

The hypothesis was wrong. The larger budget reduces *training* loss substantially — by 16%, 31%, 29% and 23% at H=96, 192, 336 and 720 — while test error does not improve anywhere (0.1401 → 0.1402, 0.1588 → 0.1612, 0.1740 → 0.1790, 0.2033 → 0.2045; dataset average 0.169 → 0.171). The train-to-validation gap widens from 14–32% to 38–69%. The optimizer is therefore not the constraint: the model can already fit the training distribution considerably harder than it can generalize, and the aggressive decay was acting as an implicit regularizer rather than starving the fit.

What limits accuracy is the generalization gap itself. Under the shipped schedule, training-to-validation error rises by 14–32% and validation-to-test error by a further 12–18% at the shorter horizons — the latter reflecting distribution shift between the validation and test periods rather than capacity. This is consistent with the two other capacity results in this paper: reducing width hurt (Table 10) and increasing optimization does not help, so the model sits near the useful capacity for this data and recipe. It also means the accuracy gap to the strongest published Electricity results is unlikely to close through tuning; it points instead at the non-stationarity handling discussed above, where normalization that adapts to the input–output distribution mismatch is the active research direction.

This experiment also strengthens the component-ablation nulls rather than qualifying them. A zero-initialized branch or mixer must *grow* its contribution, so a plausible objection was that the recipe never trained those components; under a five-fold larger budget they still do not produce a test gain, so the finding that no component effect survives correction is not an artifact of the learning-rate schedule. The training loop now emits an explicit undertrained-versus-overfit diagnostic at the end of each run so this condition is visible without inspecting curves by hand.

Statistical power and multiplicity.. With n=3 seeds, tests have limited power, and although we apply Benjamini–Hochberg correction, all 29 component-placement effects fall to $q \geq 0.11$: those results are exploratory. Raising to five and preferably ten seeds, and reporting raw p- and corrected q-values for every comparison (as we do in `stats_correction.json`), is needed to move the placement effects from directional to confirmatory. A further limitation is that our pairing is *positional*: seed index i of one configuration is paired with seed index i of another, which shares the initialization and data-shuffling seed but does not guarantee an independently verifiable common random-number stream across architectures with different parameter-sampling orders. Where such pairing cannot be assured, an unpaired or randomization-based sensitivity analysis should accompany the paired test.

Model-selection bias.. This is the most consequential threat. Although checkpoints are chosen on validation loss, some *switch* settings — most visibly disabling RevIN for Solar — were decided after inspecting test-set ablations and then folded into the reported configuration. This is not training-data leakage, but it is selection on the test signal, and it can inflate the reported numbers for the affected datasets. We therefore exclude the Solar frequency result and other test-informed placement findings from confirmatory claims. A clean follow-up must fix a candidate configuration space in advance, select *every* switch (normalization, mixer, spectral map, encoder type) on validation data only, freezes the choice before touching the test set, and reports three clearly labeled models: a single *fixed* configuration applied everywhere, a *validation-selected* model, and a *test-oracle* upper bound. Adding untouched confirmatory datasets (e.g., PEMS03/04/07/08) or previously unused temporal test blocks would then verify that the validation-only selection logic generalizes rather than having been fit to the existing nine benchmarks.

Fusion-gate initialization bias.. The convex fusion gate is biased so that $g_0 \approx 0.9$ at initialization, deliberately starting the model close to the linear temporal pathway. This is an inductive bias, not a neutral prior: it down-weights the frequency and correction pathways during the highest-learning-rate epochs and interacts with the init asymmetry noted in Section 4.2 (the spectral head is zero-init only when the FITS backbone is enabled, and randomly initialized otherwise). Both choices could depress the measured branch-removal effects, so the branch matrix should be read as a lower bound on each domain's importance under this particular initialization. A gate-bias sweep and a matched random-vs-zero head-init comparison would quantify how much of the temporal dominance we report is architectural rather than a consequence of the starting point.

Channel-order sensitivity.. The bidirectional variate mixer is non-causal but not permutation-invariant. All results use the canonical dataset column order, and no permutation stress test is available. Consequently, part of the measured mixer effect may depend on an arbitrary ordering. A permutation-equivariant mixer or repeated random-order evaluation is needed to resolve this limitation.

Coverage and efficiency.. Electricity, Traffic, and Exchange enter the branch matrix at H=96 only, and the other component ablations cover only H=96; schedule sensitivity and head-norm dynamics are not measured; and the complexity table lacks wall-clock, matched-baseline throughput, latency, and peak-memory figures. The fixed 96-step look-back leaves the Traffic weekly-cycle hypothesis untested, while Exchange at H=720 remains a notable failure case. Explicit forecasting of time-varying dispersion is left as future work rather than claimed as part of the evaluated model.

7. Conclusion

DD-Mamba is a forecast-decomposable framework built on state-space branches — a DLinear-style temporal backbone, a selective-state-space correction, an optional frequency pathway, and a switchable cross-variate mixer, each an independently removable component. On the six datasets and seven baselines evaluated in one pipeline, it obtains the lowest average MSE on five datasets; corrected seed-matched tests support four differences, while Weather is tied and PatchTST is better on ETTm1. These results are restricted to the implemented comparison and do not establish a broader state-of-the-art ranking.

The component study provides a more qualified result. None of the 29 placement effects survives multiplicity correction at three seeds. Eight of 54 branch-removal tests support the temporal pathway, with the most consistent pattern on ETTm1. The sole corrected frequency-side effect is Solar at H=96, but it is treated as exploratory because the Solar normalization switch was selected with test feedback. Zero initialization is accuracy-neutral, and PCC and dispersion summaries remain descriptive covariates rather than validated selectors. The present evidence therefore supports forecast-decomposable analysis and dataset-specific hypotheses. A general component-selection rule requires validation-only configuration, untouched datasets, stronger baseline coverage, more ablation runs, channel permutation tests, and matched efficiency measurements.

CRediT authorship contribution statement

First Author: Conceptualization, Methodology, Software, Investigation, Writing – original draft. **Second Author:** Supervision, Validation, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All nine benchmark datasets are publicly available (ETT, Weather, Solar, Electricity, Traffic, Exchange). Code, configuration files, seed-level outputs, statistical-analysis scripts, and training logs will be deposited in an anonymized repository for review: *[repository URL to be inserted before submission]*.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used Anthropic Claude in order to assist with language editing, LaTeX revision, and code scaffolding for the scientific figures. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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